

Chemistry

Unit 2

Practice Exam 2

This sample exam paper has been prepared as part of the Pearson suite of resources for the Year 11, Unit 2, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes Working time: 90 minutes

Materials required

An approved non-programmable calculator

Chemistry Data Sheet. This may be downloaded from the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	13	13	25	13	14
Section two: Short answer	6	6	30	38	41
Section three: Extended answer	4	4	35	42	45
				Total	100

Section one: Multiple-choice

14% (13 marks)

This section has 13 questions. Answer **all** questions by circling the correct option.

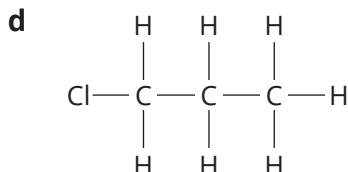
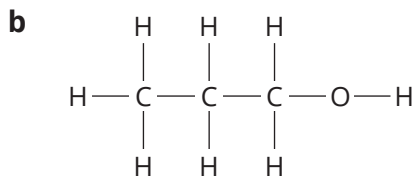
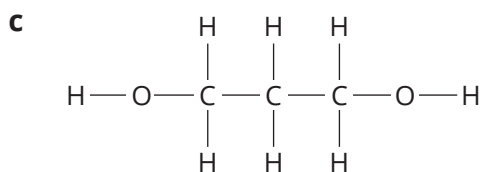
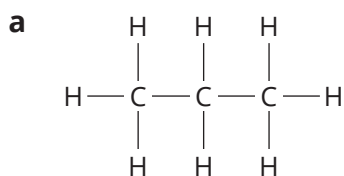
Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 25 minutes

- 1 The vapour pressure of methanol (CH_3OH) at 25°C is 16.9 kPa. At the same temperature the vapour pressure of ethanol ($\text{C}_2\text{H}_5\text{OH}$) is 7.8 kPa.

From this data we can deduce that, in comparison to ethanol,

- a the bonds between methanol molecules are stronger and the boiling point of methanol is higher.
 - b the bonds between methanol molecules are weaker and the boiling point of methanol is higher.
 - c the bonds between methanol molecules are stronger and the boiling point of methanol is lower.
 - d the bonds between methanol molecules are weaker and the boiling point of methanol is lower.
- 2 The number of lone pairs of electrons in a molecule of oxygen is
- a 2.
 - b 4.
 - c 6.
 - d 8.
- 3 Which of the following groups contains only non-polar molecules?
- a HF , O_2 , H_2 , N_2
 - b CO_2 , CH_4 , O_2 , N_2
 - c H_2 , HF , HCl , O_2
 - d N_2 , CO_2 , NH_3 , CH_4
- 4 Which one of the following compounds is most soluble in water?



- 5 The formation of a solution involves
- breaking bonds between solute particles.
 - breaking intermolecular bonds between solvent molecules.
 - the formation of bonds between solvent molecules and solute particles.
 - breaking bonds between solute particles, breaking intermolecular bonds between solvent molecules, and the formation of bonds between solvent molecules and solute particles.
- 6 In what order does the pH increase for the following substances: pure water, 0.001 M HCl, 0.001 M CH₃COOH, and 0.001 M NaOH?
- HCl < CH₃COOH < H₂O < NaOH
 - H₂O < NaOH < CH₃COOH < HCl
 - CH₃COOH < HCl < H₂O < NaOH
 - NaOH < HCl < CH₃COOH < H₂O
- 7 Which one of the following ionic equations best represents the reaction between magnesium and hydrochloric acid?
- $\text{Mg(s)} + \text{H}^+(\text{aq}) \rightarrow \text{Mg}^+(\text{aq}) + \text{H}_2(\text{g})$
 - $2\text{Mg(s)} + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Mg}^+(\text{aq}) + \text{H}_2(\text{g})$
 - $\text{Mg(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
 - $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2(\text{g})$
- 8 What volume of water must be added to 10.0 mL of NaCl to change its concentration from 0.100 M to 0.040 M?
- 15 mL
 - 25 mL
 - 40 mL
 - 250 mL
- 9 The ionic equation for the reaction that occurs when solutions of barium chloride and sodium sulfate are mixed is
- $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$.
 - $2\text{Na}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{s})$.
 - $\text{BaCl}(\text{aq}) + \text{NaSO}_4(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{BaSO}_4(\text{s})$.
 - $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{BaSO}_4(\text{s})$.
- 10 The complete combustion of propane gas can be represented by the equation
- $$\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}).$$
- What volume of carbon dioxide gas, measured at STP, would be produced when 500 g of propane was burnt in excess oxygen?
- 86 L
 - 260 L
 - 773 L
 - 1500 L

- 11** Which one of the following has the greatest mass?
- a** 0.25 mol of cobalt
 - b** 14.0 g of $(\text{NH}_4)_2\text{SO}_4$
 - c** 2.4×10^{23} carbon dioxide molecules
 - d** 98 L of helium gas at STP
- 12** One method of determining the rate of a chemical reaction is by measuring the change in mass of the reaction mixture over time. For which one of the following reactions would this method NOT be suitable? (Assume that all reactions are carried out in an open beaker.)
- a** Dilute hydrochloric acid solution added to magnesium carbonate powder.
 - b** Silver nitrate solution added to potassium chloride solution.
 - c** Hydrogen peroxide solution forming water and oxygen when a catalyst is added to it.
 - d** Magnesium metal added to a solution of hydrochloric acid.
- 13** In which of the following would the rate of reaction between 0.50 g magnesium and 2.0 M hydrochloric acid be greatest?
- a** Magnesium powder added to acid at 60°C.
 - b** Magnesium ribbon added to acid at 60°C.
 - c** Magnesium powder added to acid at 20°C.
 - d** Magnesium ribbon added to acid at 20°C.

End of section one

Section two: short answer

41% (38 marks)

This section has 6 questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 30 minutes

Question 14

(5 marks)

Ammonia (NH_3) is a constituent of many cleaning products.

- a** Draw the Lewis structure of an ammonia molecule, including non-bonding electron pairs.

(1 mark)

- b** Draw and name the shape of an ammonia molecule.

(1 mark)

- c** Briefly discuss the nature of the bonding between ammonia molecules and provide an explanation why this bonding occurs.

(3 marks)

Question 15**(4 marks)**

Polystyrene is a polymer of styrene. Styrene is made from ethylbenzene in a reaction where the conversion to styrene is a slow reaction. The mixture is boiled until the percentage of styrene exceeds 95%. Gas chromatography can be used to determine the percentage conversion to styrene.

- a** Explain how gas chromatography can be used to prove that the only two components in the reaction are ethylbenzene and styrene. (2 marks)

- b** Explain how you could use gas chromatography to experimentally determine the styrene concentration. (2 marks)

Question 16**(4 marks)**

- a** State in words Charles' law, describing the relationship between the temperature and volume of a gas. (2 marks)

- b** Use the kinetic molecular theory to explain the relationship stated in Charles' law. (2 marks)

Question 17**(9 marks)**

A student has three solutions of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$). One is unsaturated, one is saturated and one is supersaturated.

- a** Describe what is meant by the term 'unsaturated'. (1 mark)

- b** Devise and describe a simple method that could be used to determine which solution is which. Describe the results you would expect for each solution. (3 marks)

- c** A solution of sodium thiosulfate is made by dissolving 5.67 g of the solid in distilled water to make up 250.0 mL of solution. The molar mass of $\text{Na}_2\text{S}_2\text{O}_3$ is $158.12 \text{ g mol}^{-1}$.

What is the concentration of this solution in:

- i** g L^{-1} ? (1 mark)

- ii** mol L^{-1} ? (2 marks)

- iii** If this solution is added to a beaker containing 350 mL of water, what is the new concentration in mol L^{-1} ? (2 marks)

Question 18**(11 marks)**

Five solutions are known to be 0.1 M sodium hydroxide (NaOH), 0.1 M hydrochloric acid (HCl), 0.1 M glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), 0.1 M ammonia (NH_3), and 0.1 M ethanoic acid (CH_3COOH). In order to identify them, temporary labels, A–E, were placed on the bottles and the electrical conductivity and pH of each solution was measured.

- a** Using the tabulated results below, identify each solution and briefly explain your reasoning. (5 marks)

Solution	Electrical conductivity	pH
A	poor	11
B	zero	7
C	good	13
D	good	1
E	poor	3

Solution A

Solution B

Solution C

Solution D

Solution E

- b** Consider the following acid-based reactions. For each reaction, predict what you would note when the reactants are mixed and write an ionic equation for the reaction.

i Dilute hydrochloric acid is added to solid iron(II) hydroxide. (3 marks)

ii Sodium hydroxide solution is added to a solution of ammonium chloride. (3 marks)

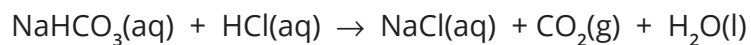
Question 19

(5 marks)

A student is provided with a solution of 0.0500 M sodium hydrogen carbonate solution and hydrochloric acid solution of unknown concentration.

- a** Calculate the mass of sodium hydrogen carbonate required to make up 200.0 mL of 0.0500 M solution. (3 marks)

- b** 23.0 mL of the sodium hydrogen carbonate solution is required to react with 35.0 mL of the hydrochloric acid solution as per the following equation.



Calculate the concentration of the hydrochloric acid.

(2 marks)

End of section two

Section three: Extended answer

45% (42 marks)

This section has 4 questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

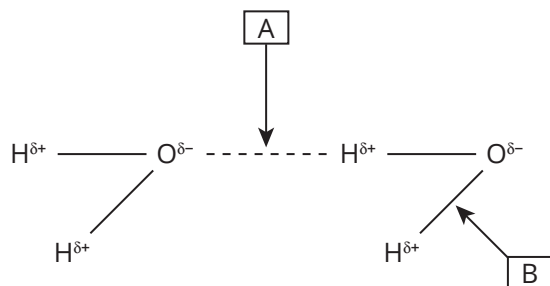
Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 35 minutes

Question 20

(8 marks)

- a** The interaction between two water molecules in a sample of water is shown below.



- i** What type of bond is A? _____ (1 mark)
- ii** What type of bond is B? _____ (1 mark)
- b** Explain, with reference to its structure and bonding, why water has a relatively high boiling temperature. (2 marks)

- c** Explain, with reference to its structure and bonding, why water expands on freezing. (2 marks)

- d** Explain, with reference to structure and bonding, why many ionic and polar substances will dissolve in water. (2 marks)

Question 21**(12 marks)**

A compound composed only of carbon and hydrogen contains 80% by mass of carbon.

- a** Determine the empirical formula of the compound. (3 marks)

- b** At STP, 6.72 g of the compound occupies 5.15 L. Determine the molar mass, in g mol^{-1} , of the compound. (2 marks)

- c** Determine the molecular formula of the gas. (3 marks)

- d** Identify this compound and draw its molecular structural formula. (2 marks)

Name of compound _____

Structural formula

- e** Identify the type of intermolecular forces that exist between molecules of this compound and briefly describe how these forces are formed. (2 marks)

Question 22**(12 marks)**

You are given five solutions. They are not labelled but are known to be sodium carbonate, sulfuric acid, potassium nitrate, magnesium nitrate and copper(II) nitrate. You have been set the task to identify the solutions.

- a** In the grid below, predict what you would **observe** if a sample of each solution was added to a sample of each of the other four solutions. (If no observable reaction is predicted, write 'no change'.) (4 marks)

	Sodium carbonate	Sulfuric acid	Potassium nitrate	Magnesium nitrate	Copper(II) nitrate
Sodium carbonate					
Sulfuric acid					
Potassium nitrate					
Magnesium nitrate					
Copper(II) nitrate					

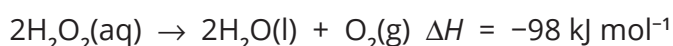
- b** Write a balanced ionic equation for each of the reactions predicted in part a. (3 marks)

- c** Describe how the solutions can be identified by mixing each solution with each of the other four solutions. (5 marks)

Question 23**(10 marks)**

Hydrogen peroxide (H_2O_2) has a wide range of uses, both in the home and in industry. A 3% solution is used as a disinfectant for the treatment of bacterial and fungal infections. It is a bleaching agent used in some hair bleaches and laundry detergents. A concentrated solution of hydrogen peroxide is used to bleach wood pulp for the paper industry. It is used in waste water treatment and in the manufacture of explosives. It is even used as a rocket propellant!

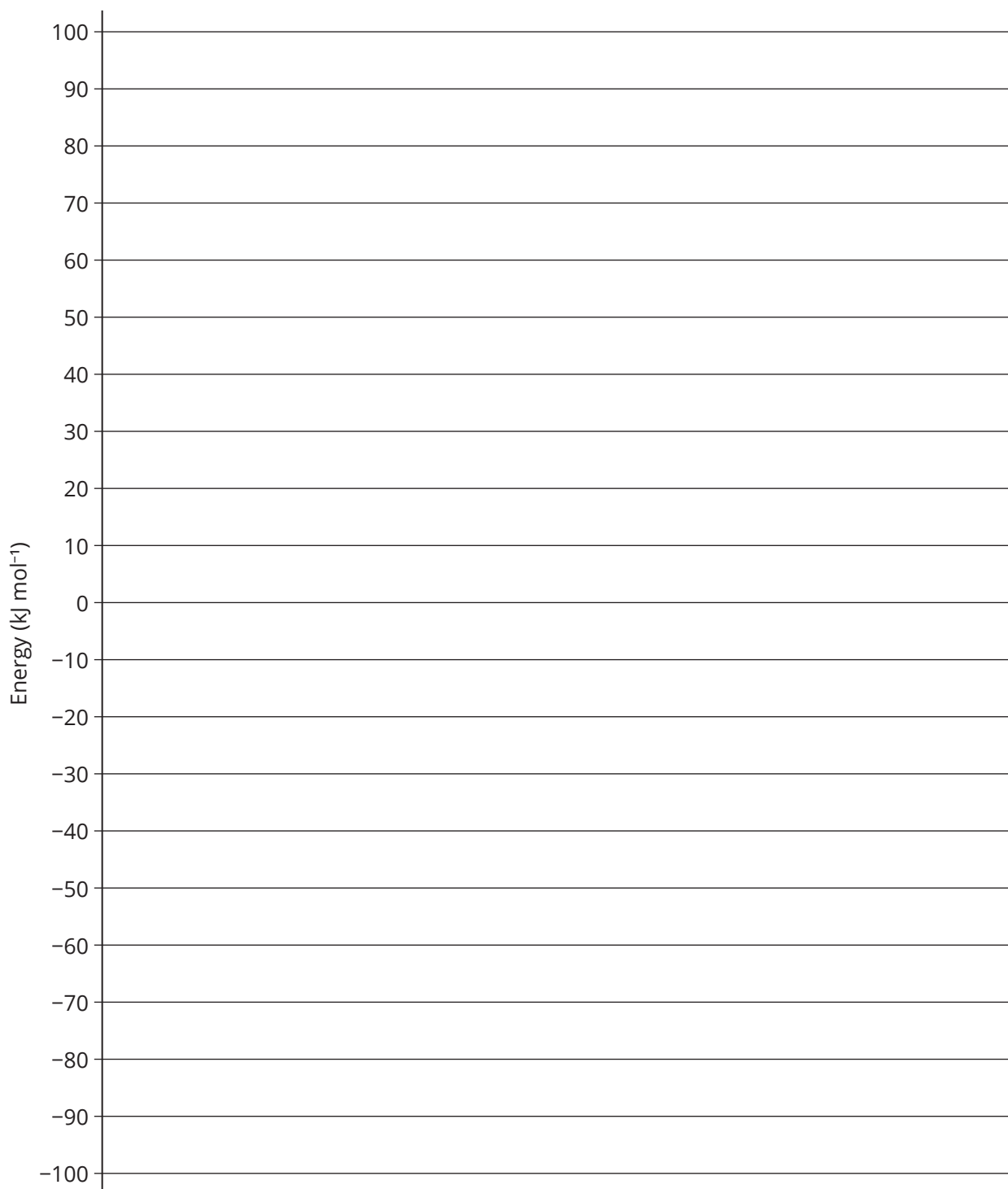
Hydrogen peroxide is unstable and decomposes to form oxygen and water.



At 300 K the activation energy for this decomposition reaction is 75 kJ mol^{-1} . When an iodine catalyst is used at the same temperature the activation energy is 56 kJ mol^{-1} .

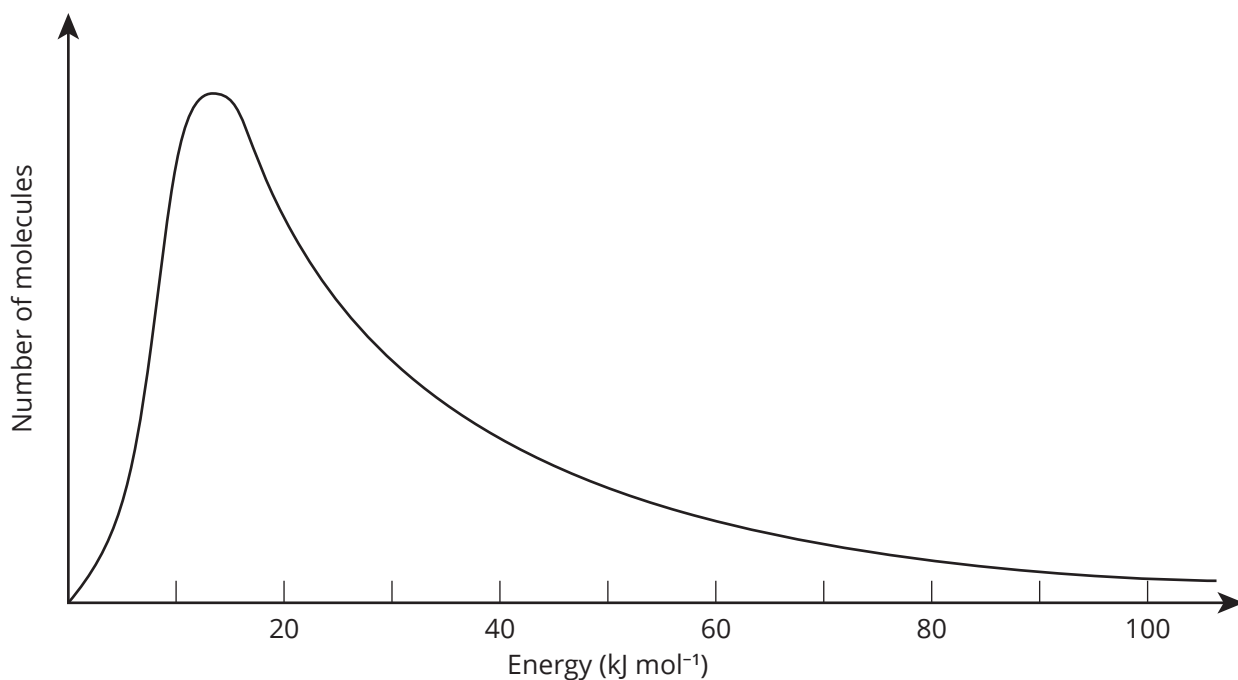
- a** Explain what is meant by the term 'activation energy'. (1 mark)

- b** On the grid below, draw an energy profile diagram for this reaction. Include both the catalysed and uncatalysed reactions. Label the appropriate activation energies and the heat of reaction. For simplicity, assume that the energy of the reactants is at an arbitrary zero. (3 marks)



- c** Explain why the addition of an iodine catalyst increases the rate of this reaction. (2 marks)

d The energy distribution graph for this reaction at 300°C is shown below.



- i** Shade in the part of the graph that represents the number of particles, in the catalysed reaction, that have energies equal to or greater than the activation energy. (1 mark)
- ii** On this graph draw a new energy distribution curve for the reaction at 400 K. (1 mark)
- iii** Explain, in terms of collision theory, why the rate of this reaction increases when the temperature is raised. (2 marks)

End of questions